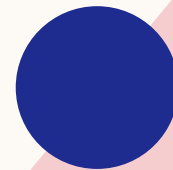


The background features a large white circle in the center, partially overlapping a light blue area on the left and a light pink area on the right. A dark blue shape is at the bottom, also overlapping the white circle. The text is centered within the white circle.

DATA PREPARATION

DATA PREPARATION



INTRODUCTION

Data preparation is the process of gathering, combining, structuring and organizing data so it can be used in business intelligence (BI), analytics and data visualization applications. The components of data preparation include data preprocessing, profiling, cleansing, validation and transformation; it often also involves pulling together data from different internal systems and external sources.

PURPOSES OF DATA PREPARATION

One of the primary purposes of data preparation is to ensure that raw data being readied for processing and analysis is accurate and consistent so the results of BI and analytics applications will be valid. Data is commonly created with missing values, inaccuracies or other errors, and separate data sets often have different formats that need to be reconciled when they're combined. Correcting data errors, validating data quality and consolidating data sets are big parts of data preparation projects.

WHAT ARE THE BENEFITS OF DATA

PREPARATION?

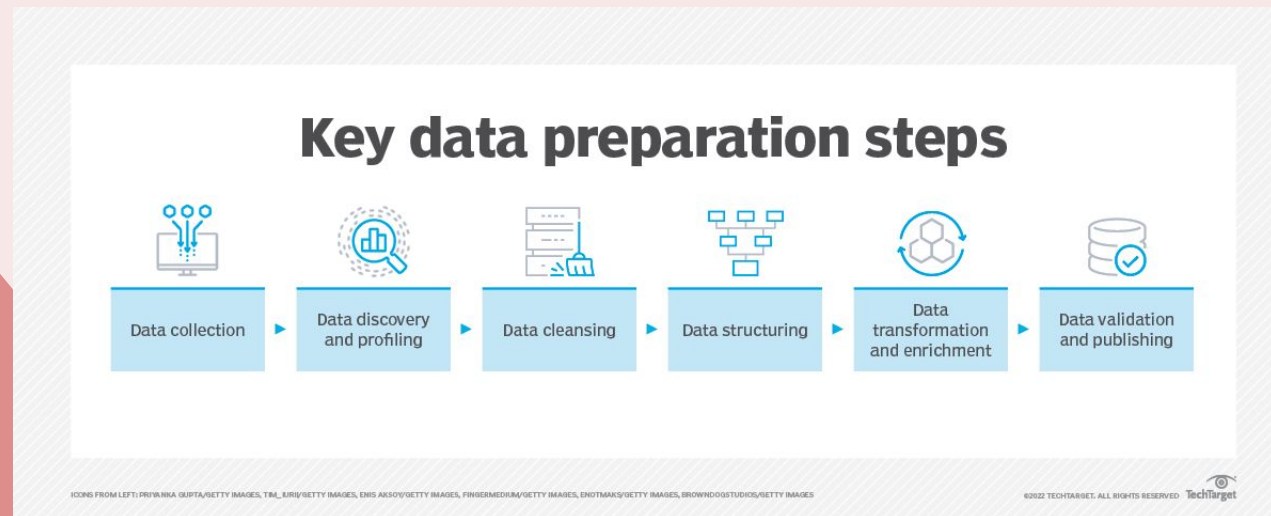
Data scientists often complain that they spend most of their time gathering, cleansing and structuring data instead of analyzing it. A big benefit of an effective data preparation process is that they and other end users can focus more on data mining and data analysis -- the parts of their job that generate business value. For example, data preparation can be done more quickly, and prepared data can automatically be fed to users for recurring analytics applications.

Done properly, data preparation also helps an organization do the following:

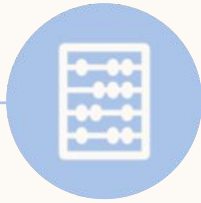
- ensure the data used in analytics applications produces reliable results;
- identify and fix data issues that otherwise might not be detected;
- enable more informed decision-making by business executives and operational workers;
- reduce data management and analytics costs;
- avoid duplication of effort in preparing data for use in multiple applications; and
- get a higher ROI from BI and analytics initiatives.

STEPS IN THE DATA PREPARATION PROCESS

Data preparation is done in a series of steps. There's some variation in the data preparation steps listed by different data professionals and software vendors, but the process typically involves the following tasks:



STEPS IN THE DATA PREPARATION PROCESS



DATA COLLECTION.

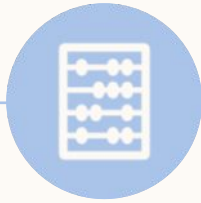
Relevant data is gathered from operational systems, data warehouses, data lakes and other data sources. During this step, data scientists, members of the BI team, other data professionals and end users who collect data should confirm that it's a good fit for the objectives of the planned analytics applications.



DATA DISCOVERY AND PROFILING

The next step is to explore the collected data to better understand what it contains and what needs to be done to prepare it for the intended uses. To help with that, data profiling identifies patterns, relationships and other attributes in the data, as well as inconsistencies, anomalies, missing values and other issues so they can be addressed.

STEPS IN THE DATA PREPARATION PROCESS



DATA CLEANSING

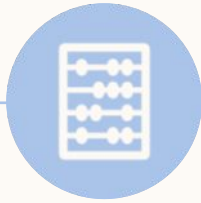
Next, the identified data errors and issues are corrected to create complete and accurate data sets. For example, as part of cleansing data sets, faulty data is removed or fixed, missing values are filled in and inconsistent entries are harmonized.



DATA STRUCTURING

At this point, the data needs to be modeled and organized to meet the analytics requirements. For example, data stored in comma-separated values (CSV) files or other file formats has to be converted into tables to make it accessible to BI and analytics tools.

STEPS IN THE DATA PREPARATION PROCESS



DATA TRANSFORMATION AND ENRICHMENT

In addition to being structured, the data typically must be transformed into a unified and usable format. For example, data transformation may involve creating new fields or columns that aggregate values from existing ones. Data enrichment further enhances and optimizes data sets as needed, through measures such as augmenting and adding data.



DATA VALIDATION AND PUBLISHING

In this last step, automated routines are run against the data to validate its consistency, completeness and accuracy. The prepared data is then stored in a data warehouse, a data lake or another repository and either used directly by whoever prepared it or made available for other users to access.

WHAT ARE THE CHALLENGES OF DATA PREPARATION?

Data preparation is inherently complicated. Data sets pulled together from different source systems are highly likely to have numerous data quality, accuracy and consistency issues to resolve. The data also must be manipulated to make it usable, and irrelevant data needs to be weeded out. As noted above, it's a time-consuming process: The 80/20 rule is often applied to analytics applications, with about 80% of the work said to be devoted to collecting and preparing data and only 20% to analyzing it.

Top challenges on data preparation



- Inadequate or nonexistent data profiling
- Missing or incomplete data
 - Invalid data values
- Name and address standardization
- Inconsistent data across enterprise systems
- Data enrichment
- Maintaining and expanding data prep processes

WHAT ARE THE CHALLENGES OF DATA PREPARATION?

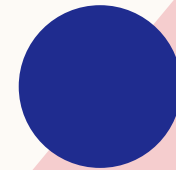
In an article on common data preparation challenges, Rick Sherman, managing partner of consulting firm Athena IT Solutions, detailed the following seven challenges along with advice on how to overcome each of them:

- **Inadequate or nonexistent data profiling.** If data isn't properly profiled, errors, anomalies and other problems might not be identified, which can result in flawed analytics.
- **Missing or incomplete data.** Data sets often have missing values and other forms of incomplete data; such issues need to be assessed as possible errors and addressed if so.
- **Invalid data values.** Misspellings, other typos and wrong numbers are examples of invalid entries that frequently occur in data and must be fixed to ensure analytics accuracy.

WHAT ARE THE CHALLENGES OF DATA PREPARATION?

- **Name and address standardization.** Names and addresses may be inconsistent in data from different systems, with variations that can affect views of customers and other entities.
- **Inconsistent data across enterprise systems.** Other inconsistencies in data sets drawn from multiple source systems, such as different terminology and unique identifiers, are also a pervasive issue in data preparation efforts.
- **Data enrichment.** Deciding how to enrich a data set -- for example, what to add to it -- is a complex task that requires a strong understanding of business needs and analytics goals.
- **Maintaining and expanding data prep processes.** Data preparation work often becomes a recurring process that needs to be sustained and enhanced on an ongoing basis.

PREPARE YOUR DATA FOR ANALYSIS USING EXCEL



INTRODUCTION

1. Import data

- ✓ Split data along delimiters (e.g. comma or semicolon)
- ✓ Extract parts from data entries
- ✓ Remove leading and trailing spaces

2. Format adjustments

- ✓ Standardize formats (e.g. dates, currencies, units)
- ✓ Store data in the correct format (e.g. value, text, date)
- ✓ Replace unrecognized or corrupted characters (e.g. ÃŸ)
- ✓ Check for truncated entries (i.e. entries that have been cut off)

3. Correct inconsistencies

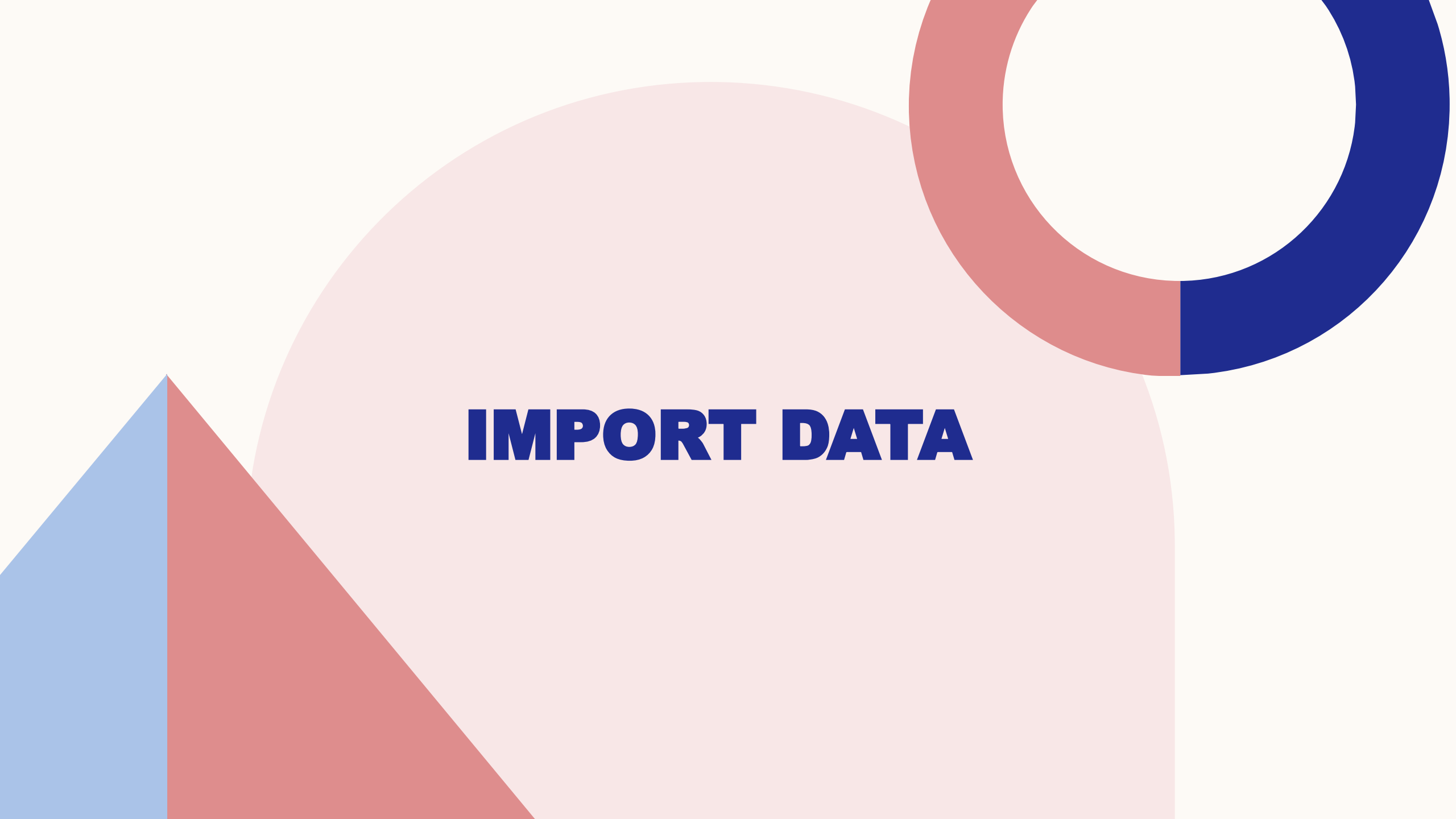
- ✓ Check for inconsistent entries using custom rules (e.g. age > 0)
- ✓ Numerical data: Check for outliers
- ✓ Survey data: Recognize suspicious response patterns
- ✓ Categorical data: Check for wrong categories (e.g. „no” vs. „0”)
- ✓ Missing values: Add data or remove rows

4. Remove duplicates

- ✓ Deduplicate your data considering fuzzy duplicates

5. Combine data sets

- ✓ Lookup values from other tables

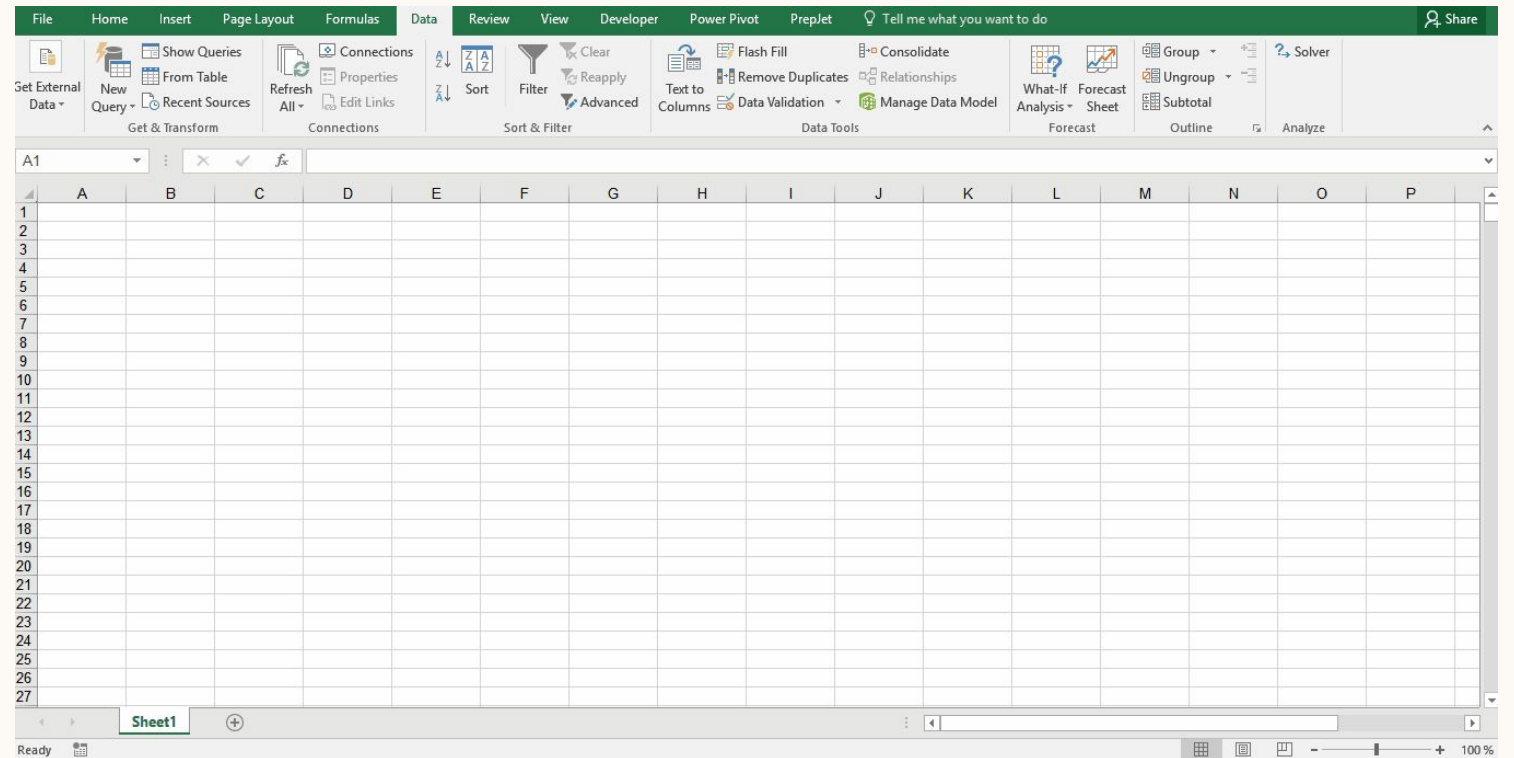


IMPORT DATA

- **SPLIT DATA ALONG DELIMITERS**

When you import your data, you should be aware of a clean delimitation of your entries. IT systems usually define a delimiter such as a semicolon or a comma. In Microsoft Excel, we recommend to use the import function to get your data into a tabular shape, also if you import a csv and not a txt file. Compared to simply opening the csv file, the import function has the advantage that you can define the character encoding. Our animation shows you how to use the import function.

- # SPLIT DATA ALONG DELIMITERS



- **EXTRACT PARTS FROM DATA ENTRIES**

In case your import did not work well or you need only a part of a certain attribute (such as a part from a nested ID), you have to perform more advanced split operations. For example, if you want to extract the domains of email addresses, you have to extract the part between the @ and the last dot. If you use original Excel functions, LEFT, RIGHT will help you to extract text parts. With FIND, you can specify the delimiters (in our case “@” and “.”), with LEN, you can define the length of the character sequences you want to extract. You will have to workaroud with a few combinations to finally get the domain:

- EXTRACT PARTS FROM DATA ENTRIES**

=LEFT(RIGHT(E2;LEN(E2)-FIND("@";E2));FIND(".",RIGHT(E2;LEN(E2)-FIND("@";E2)))-1)

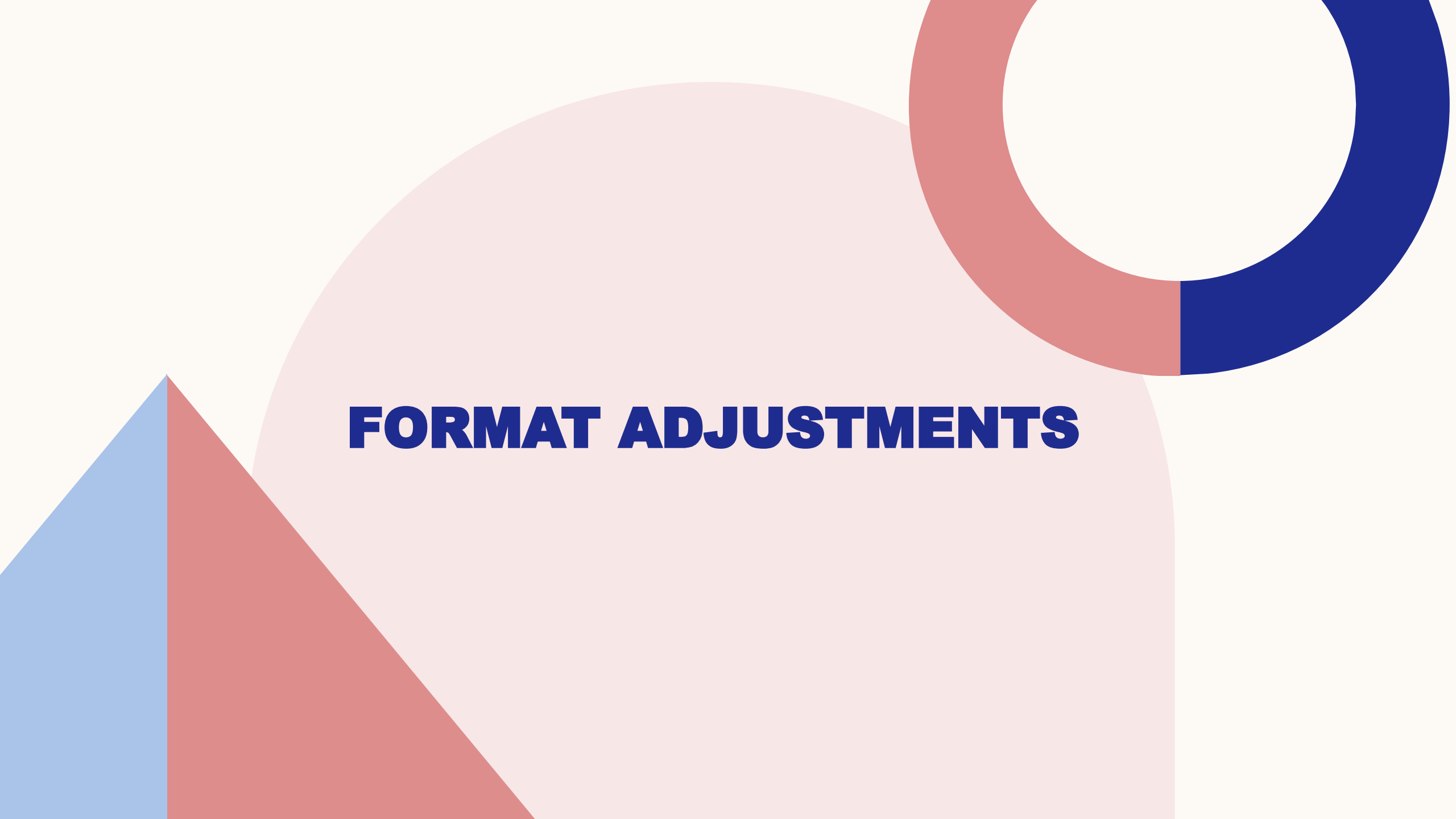
E	F	G	H	I
email	domain			
jbaker0@symantec.com	symantec			
cweaver1@ebay.co.uk	ebay			
tbaker2@wikipedia.org	wikipedia			
jgarza3@boston.com	boston			
swright4@netlog.com	netlog			
rhanson5@angelfire.com	angelfire			
pgordon6@whitehouse.gov	whitehouse			
abrooks7@biblegateway.com	biblegateway			
vstanley8@craigslist.org	craigslist			
gwatkins9@instagram.com	instagram			
cgranta@mediafire.com	mediafire			
wgriffinb@alexa.com	alexa			
asimsc@oakley.com	oakley			
bfernandezd@addthis.com	addthis			
kportere@toplist.cz	toplist			

- **REMOVE LEADING AND TRAILING SPACES**

System exports or web-scraped data often cause unwanted leading and trailing spaces. You can remove those spaces by using the Excel TRIM function (do not use find and replace as this will also remove spaces between words!). The syntax is as follows:

=TRIM(text)

If you do not want to repeat this exercise for every single column, you can use PrepJet's Trim Spaces function which allows you to trim your whole sheet in a single click.



FORMAT ADJUSTMENTS

• **STANDARDIZE FORMATS**

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Before combining and analyzing data, it is crucial to harmonize the formats of your data. If your data comes from different countries or IT systems with different languages, you should make sure to have consistent decimal separators (comma vs. dot). The same applies for date formats (e.g. DD.MM.YYYY vs. MM/DD/YYYY) currencies (e.g. EUR vs. USD) or measurement units (e.g. miles vs. kilometers). In Excel, the best solution is to transform data sets with different formats into one standard before combining them. For date types, you can use Excel's "Number" functionality. Click on the lower right arrow to define a custom format:

• STANDARDIZE FORMATS

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The screenshot shows the Microsoft Excel interface with the 'Format Cells' dialog box open. The 'Number' tab is selected, and the 'Date' category is chosen. The 'Type' list shows various date formats, with '14-Mar-12' selected. The 'Locale (location)' is set to 'English (United States)'. An orange arrow points to the 'Number' dropdown in the ribbon, which is currently set to 'General'.

Format Cells

Category: General, Number, Currency, Accounting, **Date**, Time, Percentage, Fraction, Scientific, Text, Special, Custom

Sample: 15.03.2023

Type: 3.14, 3.14.12, 03.14.12, 14-Mar, **14-Mar-12**, 14-Mar-12, Mar-12

Locale (location): English (United States)

Date formats display date and time serial numbers as date values. Date formats that begin with an asterisk (*) respond to changes in regional date and time settings that are specified for the operating system. Formats without an asterisk are not affected by operating system settings.

OK Cancel

- **STORE DATA IN THE CORRECT FORMAT**

To make sure that your data is analyzed appropriately, you should store it in the correct format. For example, your data might contain a numerical Identifier which has however no numerical meaning. Tell Excel that this is not a number by classifying it as text. Excel helps you to specify the date format in the Home ribbon in the section “Number” (see above). To perform a final check on the format of your data, you can use PrepJet’s Find Inconsistencies function which highlights inconsistent data types automatically.

- **REPLACE UNRECOGNIZED OR CORRUPTED CHARACTERS**

In case your import failed for some reason, some symbols might be corrupted after an export as the encoding of characters might differ between IT systems. Characters like “ÃŸ” might introduce problems if you want to work with your data. We recommend to correct them using the find and replace function.

- **CHECK FOR TRUNCATED ENTRIES**

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When exporting data from IT systems, a lot of annoying accidents can happen. One of them is truncations, i.e. data entries are cut off at a certain position. Some manual screening should quickly lead you to suspicious data. How to fix it? The best solution is probably to request a new and healthy export.



**CORRECT
INCONSISTENCIES**

• CHECK FOR TRUNCATED ENTRIES

The most valuable resource when preparing data for analysis is your own knowledge about the data. With custom rules (so-called Business Rules), you can detect wrong data. The whole process follows the principles of Boolean logic . You can define simple conditions. Before applying any rules to your data, it is always helpful to spell them in natural language, e.g.: “The entries in column ‘Age’ always have to be greater or equal zero.”

PrepJet has a functionality that helps you to implement custom validation rules easily without a lot of nested formulas and additional validation columns. You can also implement validation rules in Excel using the IF function. You will have an additional validation column that indicates whether the rule is fulfilled or not (this is also the column into which you type your formula). Here, we want the output in our validation column to be “ok” if the Age is greater or equal zero and we want to have “error” if the Age is smaller than zero (assuming that column F contains Age):

=IF(F2>=0; “ok”; “error”)

• **CHECK FOR TRUNCATED ENTRIES**

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You can also implement more complex conditions. For example, if you want to analyze your product range, you could apply the rule “The shipping weight of an article of the category pants is always smaller than 2kg.” To make the implementation in Excel easier, it is helpful to phrase the condition closer to logical terminology first:

IF “category” EQUAL TO “pants” THEN “shipping_weight” SMALLER THAN “2”

In Excel, we again have a validation column that indicates whether our rule is violated (“error”) or not (“ok”).

To be prepared for Excel’s syntax, we have to rephrase the logical statement as follows:

- **CHECK FOR TRUNCATED ENTRIES**

IF “category” EQUAL TO “pants” AND “shipping_weight” SMALLER THAN “2”
THEN “ok” ELSE “error”

Assuming that column B contains category and column C contains shipping weight, we insert the following formula into the validation column and drag it down to the end of our table:

=IF(AND(B2=”pants”; C2<2); “ok”; ”error”)

You can even go further and introduce conditions with more nestings, conditions that have to be fulfilled cumulatively (using the AND function) or conditions that can be fulfilled selectively (using the OR function). Again, you can make your life a little easier using PrepJet’s Rule-based Validation function.

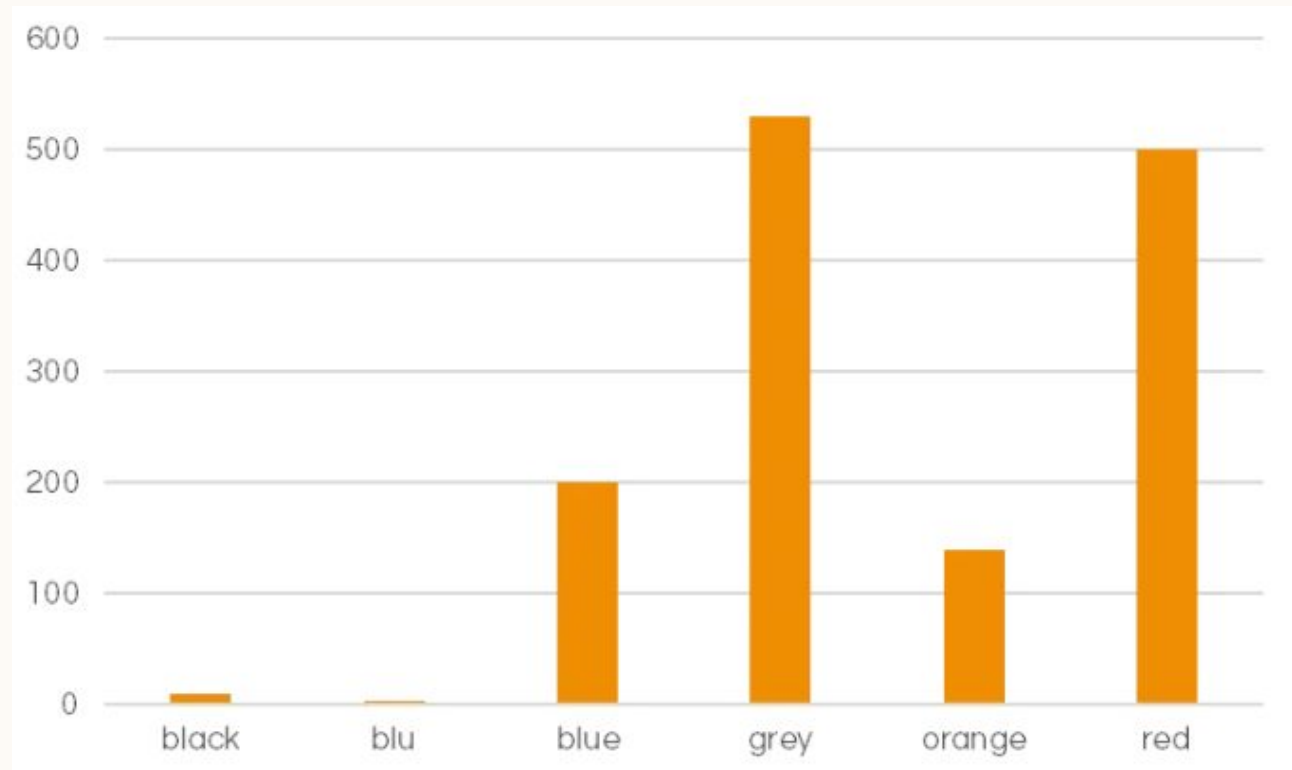
- **NUMERICAL DATA: CHECK FOR OUTLIERS**

If you work with numerical data, you should check your data for outliers. Outliers are values that deviate from the observed distribution of your data (learn more in our blog post on the basics of outlier detection). As statistical outlier detection is rather complex to be implemented in Excel, we recommend to sort the values by size and to check if there are any suspicious entries at the upper or lower bounds of your range.

- **CATEGORICAL DATA:
CHECK FOR WRONG
CATEGORIES**

If you analyze categorical data, it is important to make sure that there are no different conventions for assigning categories. For example, if you want to analyze your product portfolio, make sure that similar products are not put into different categories (e.g. if you work with a grocery store, all apples, bananas and mangos might be classified as “fruits” while someone accidentally tagged pineapple as “exotic fruit”. There are different ways to detect those kind of miscategorizations in Excel. If you have a small number of categories, you can simply insert a filter and manually check all available categories. If you have a larger range of categories, it is more advisable to plot the frequency of occurrences in each categories (e.g. with a bar chart in Excel). You can focus on categories with low frequencies of occurrence. Here in our example chart, in a few cases “blu” has been entered instead of “blue”. You can easily fix this by using Find and Replace

- **CATEGORICAL DATA:
CHECK FOR WRONG
CATEGORIES**

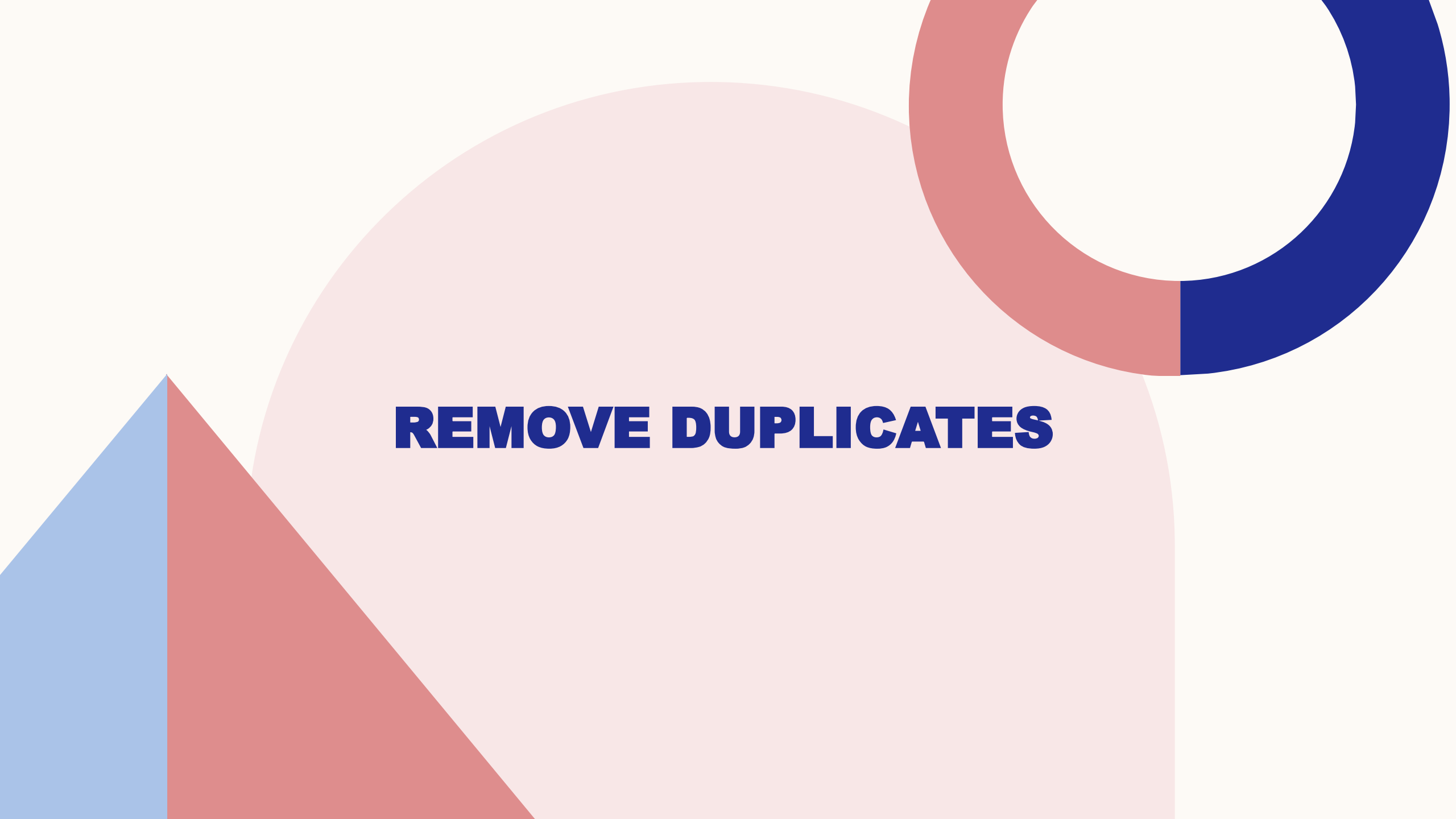


- **MISSING ENTRIES: ADD DATA OR REMOVE ROWS**

Missing values are an issue that is hard to fix. If you have the chance to get the original data without inappropriate efforts, the best solution is to add it. Otherwise, you might want to ignore the rows that contain empty cells when performing certain analyses. A more advanced option is to impute the data (e.g. by the mean or median or by a logistic regression). This is only recommendable for users with advanced statistical knowledge as it might bias your results.

- **SURVEY DATA: DETECT SUSPICIOUS RESPONSE PATTERNS**

When analyzing survey data, you should be aware of response patterns that are not valid (e.g. because the respondent simply was too tired to answer all your questions appropriately ...). This might be for example a person who always chooses the answer in the same position for each question — so-called straight-lining (e.g. always the most left option). More creative respondents also create alternating patterns when answering surveys (so-called “Christmas tree behavior”). If you use an online survey tool, as a first step you can try to identify those answers that have been filled in in very short time by looking at the response timers. Most online survey tools will export the time the respondents take to fill in the form with your survey data. In a second step, you can identify suspicious data by screening the replies in Excel. This will be easier once you translated categorical text responses into numbers (e.g. “I fully agree” = 1, “I agree” = 2 etc.).



REMOVE DUPLICATES

- **DEDUPLICATE YOUR DATA³⁷
CONSIDERING FUZZY
DUPLICATES**

Another important step before you combine and analyze data sets is to remove duplicate entries. This is simple if you only want to find exact duplicates. You can directly remove them with Excel's Remove Duplicates function in the Data Ribbon. If you first want to check the duplicate entries to examine where they come from, you can highlight and sort them with PrepJet's Detect Duplicates function.

Often, however, actual duplicates are not exactly equal. They might differ slightly due to typos or different naming conventions. To detect these kind of duplicates, you either need a sophisticated fuzzy matching algorithm or a manual workaround (learn more in our [introduction to fuzzy matching](#)). One manual solution can be to include only a few attributes into your duplicate screening. For example, if you want to find duplicates in a customer list and names might have been spelled differently, you can in a separate step only have a look at the address fields. It should also help to standardize categorical data (see above) first before searching for duplicates.



COMBINE DATA SETS

- **LOOKUP DATA FROM OTHER TABLES**

If you do not only want to analyze a single data set, the final step to get your data ready for analysis is to combine it. You could simply copy columns from one sheet to another if your data is sorted. However, this is not recommendable as you risk to copy the wrong data in case one data set misses a row. It is better to identify match criteria and pull your data based on these criteria from one table into another. If you have one match criterion that is unique for each row of your data set (a so-called unique identifier), you can use this match criterion and perform a VLOOKUP in Excel. If you have more than one match criteria, you have to use a combination of INDEX and MATCH (check out our [blog post](#) on this operation). Or you can use PrepJet's Lookup Data function, which allows you to specify as many match criteria as you want in only a few clicks.